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TRACTION BATTERY PACK CHARGING AND DISCHARGING OPERATION

Abstract

A traction battery pack operating method includes charging a traction battery pack of an electrified vehicle. The charging includes charging a first subpack of battery cells together with a second subpack of battery cells. The method can discharge the first subpack separately from the second subpack. The method can include during the discharging, maintaining at least one switch in an open state to electrically isolate the first subpack from the second subpack.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates generally to a traction battery pack and, more particularly, to charging and discharging the traction battery pack.

BACKGROUND

[0002] Electrified vehicles differ from conventional motor vehicles because electrified vehicles can be selectively driven by one or more electric machines that are powered by a traction battery pack. The electric machines can propel the electrified vehicles instead of, or in combination with, an internal combustion engine. The traction battery pack is discharged when powering the one or more electric machines and other loads of the electrified vehicle.

SUMMARY

[0003] In some aspects, the techniques described herein relate to a traction battery pack operating method, including: charging a traction battery pack of an electrified vehicle, the charging including charging a first subpack of battery cells together with a second subpack of battery cells; and discharging the first subpack separately from the second subpack.

[0004] In some aspects, the techniques described herein relate to a method, wherein the first subpack of battery cells is a first subpack of battery cells having lithium metal anodes, wherein the second subpack of battery cells is a second subpack of battery cells having lithium metal anodes.

[0005] In some aspects, the techniques described herein relate to a method, further including, during the discharging, maintaining at least one switch in an open state to electrically isolate the first subpack from the second subpack.

[0006] In some aspects, the techniques described herein relate to a method, further including, when discharging the first subpack separately from the second subpack, providing electrical power from the first subpack to a load, and electrically isolating the second subpack from the load.

[0007] In some aspects, the techniques described herein relate to a method, further including transitioning at least one switch to electrically isolate the second subpack from the load.

[0008] In some aspects, the techniques described herein relate to a method, wherein the load is at least one motor of the electrified vehicle.

[0009] In some aspects, the techniques described herein relate to a method, further including switching from discharging the first subpack separately from the second subpack, to discharging the second subpack separately from the first subpack.

[0010] In some aspects, the techniques described herein relate to a method, further including pre-charging the second subpack during the switching.

[0011] In some aspects, the techniques described herein relate to a method, wherein the discharging is during a drive cycle.

[0012] In some aspects, the techniques described herein relate to a method, further including switching based on a state of health in the first subpack, the second subpack, or both.

[0013] In some aspects, the techniques described herein relate to a method, further including switching based on a drive mode of the electrified vehicle.

[0014] In some aspects, the techniques described herein relate to a method, further including switching based on a range for the electrified vehicle when discharging from the first subpack reaching a threshold range.

[0015] In some aspects, the techniques described herein relate to a method, wherein the discharging the first subpack separately from the second subpack includes discharging the first subpack to a load without discharging the second subpack to the load.

[0016] In some aspects, the techniques described herein relate to a method, further including dividing the traction battery pack into the first subpack of battery cells, and the second subpack of

battery cells, wherein dividing the traction battery pack into the first subpack and the second subpack includes electrically isolating the second subpack from both the first subpack and a load.
[0017] In some aspects, the techniques described herein relate to a method, wherein the traction battery pack is an 800 Volt traction battery pack, wherein the first subpack is 400 Volts and the second subpack is 400 Volts.
[0018] In some aspects, the techniques described herein relate to a method, further including dividing the traction battery pack into the first subpack of battery cells, and the second subpack of battery cells, wherein the dividing further includes dividing the traction battery pack into at least one third subpack, and the discharging is a discharging of the first subpack without discharging the second subpack or the at least one third subpack.
[0019] In some aspects, the techniques described herein relate to a method, wherein the first subpack and the second subpack are both held within a battery enclosure.
[0020] In some aspects, the techniques described herein relate to a traction battery assembly, including: a traction battery pack that powers a load; and a switching system that is configured to transition between a charging state, a first discharging state, and a second discharging state, when the switching system is in charging state, a first subpack of battery cells in the traction battery pack and a second subpack of battery cells in the traction battery pack are both configured to be charged, when the switching system is in first discharging state, the first subpack is electrically connected to the load while the second subpack is electrically isolated from the load, when the switching system is in the second discharging state, the second subpack is electrically connected to the load while the first subpack is electrically isolated from the load.
[0021] In some aspects, the techniques described herein relate to a traction battery assembly, wherein the switching system electrically isolates the first subpack of battery cells from the second subpack of battery when the switching system is in the first discharging state and when the switching system is in the second discharging state.
[0022] In some aspects, the techniques described herein relate to a traction battery assembly, wherein the first subpack of battery cells is a first subpack of battery cells having lithium metal anodes, wherein the second subpack of battery cells is a second subpack of battery cells having lithium metal anodes.
[0023] The embodiments, examples and alternatives of the preceding paragraphs, the claims, or the following description and drawings, including any of their various aspects or respective individual features, may be taken independently or in any combination. Features described in connection with one embodiment are applicable to all embodiments, unless such features are incompatible.

Description

BRIEF DESCRIPTION OF THE FIGURES
[0024] The various features and advantages of the disclosed examples will become apparent to those skilled in the art from the detailed description. The figures that accompany the detailed description can be briefly described as follows:
[0025] FIG. **1** illustrates a side view of an electrified vehicle according to an exemplary aspect of the present disclosure.
[0026] FIG. **2** illustrates an expanded view of a traction battery pack from the electrified vehicle of FIG. **1**.
[0027] FIG. **3** illustrates a flow of an example method of operating the traction battery pack of FIG. **2**.
[0028] FIG. **4** illustrates a schematic view of the traction battery pack of FIG. **2** when operating in a charging state where a first subpack of the traction battery pack and a second subpack of the traction battery pack are charged together.

[0029] FIG. **5** illustrates a schematic view of the traction battery pack of FIG. **2** when operating in a first discharging state where the first subpack is powering a load and the second subpack is isolated from both the load and the first subpack.
[0030] FIG. **6** illustrates a schematic view of the traction battery pack of FIG. **2** when operating in a second discharging state where the second subpack is powering a load and the first subpack is isolated from both the load and the second subpack.

DETAILED DESCRIPTION

[0031] A traction battery pack of an electrified vehicle can include battery cells that are charged and discharged. The battery cells can be charged from, for example, an external power source or, if the electrified vehicle is a hybrid electric vehicle, from a generator of the electrified vehicle. The battery cells can be discharged to power one or more traction motors or other loads of the electrified vehicle. Charging and discharging at different rates can prolong a life of some types of battery cells, such as battery cells having lithium metal anodes.
[0032] With reference to FIGS. **1** and **2**, an electrified vehicle **10**, in an exemplary non-limiting embodiment, includes a traction battery pack **14** that powers an electric machine **18**. The electrified vehicle **10** further includes wheels **22** driven by the electric machine **18**. The traction battery pack **14** can power the electric machine **18**, which converts electric power to torque to drive the wheels **22**.
[0033] The electrified vehicle **10** is an all-electric vehicle. In other examples, the electrified vehicle **10** is a hybrid electric vehicle, which can selectively drive wheels using torque provided by an internal combustion engine instead, or in addition to, an electric machine. Generally, the electrified vehicle **10** could be any type of vehicle having a traction battery pack.
[0034] The traction battery pack **14** includes an enclosure **30** that encloses a plurality of battery arrays **34**. Each of the battery arrays **34** includes a plurality of individual battery cells **38**, which, in this example, have lithium metal anodes. Such battery cells **38** can provide relatively high energy densities and can have their cycle lives extended by discharging at rates significantly higher than the rate at which they are charged. The battery cells with the lithium metal anodes can include battery cells with liquid electrolytes, battery cells with solid electrolytes, or some combination thereof.
[0035] With reference now to FIGS. **3-6** and continuing reference to FIGS. **1** and **2**, a method **100** of operating the traction battery pack **14** generally includes a step **104** of charging the traction battery pack **14**. The step **104** can include charging the traction battery pack **14** from an power source **40**, which can be an external power source, such as a charge station. In another example, the charging is from one or more generators of the electrified vehicle **10**.
[0036] The charging includes charging a first subpack **62** of battery cells **38** together with a second subpack **64** of battery cells **38**. The first subpack **62** and the second subpack **64** are both held within the enclosure **30** in this example.
[0037] In this example, the first subpack **62** can include two of the battery arrays **34**. The second subpack **64** can include the two remaining battery arrays **34**. If the traction battery pack **14** is an 800 Volt pack, the first subpack **62** and the second subpack **64** can then each be 400 Volts. Packs having other voltages could be used in other examples, including 1200 Volt packs.
[0038] The charging of the first subpack **62** and the second subpack **64** can occur when the first subpack **62** and the second subpack **64** are connected in parallel or in series. Charging when the first subpack **62** and the second subpack **64** are connected in series as an 800 Volt pack can lower the current required to meet a charging power goal.
[0039] In this example, the battery cells **38** of the first subpack **62** and the battery cells **38** of the second subpack **64** have the same chemistries (i.e., both nickel manganese cobalt (NMC), lithium iron phosphate (LFP), or nickel cobalt aluminum (NCA). In other examples, the battery cells **38** and the battery cells **38** of the second subpack **64** have different chemistries. The chemistries need not be the same, but the voltage scale between the first subpack **62** and the second subpack **64** is

typically similar.
[0040] A switching system **70** is transitioned to electrically connect together the first subpack **62** and the second subpack **64** during charging. The switching system **70**, in this example, includes switches S**1**, S**2**, S**3**, and S**4**. Switches S**1** and S**2** are transitioned to a closed state during charging. Switches S**3** and S**4** are transitioned to an open state.
[0041] In an embodiment, a control module **74**, such as a vehicle controller, executes the method **100** by controlling the transitions of the switching system **70**. However, other configurations and types of control are contemplated within the scope of this disclosure. Although shown schematically as a single control module **74**, a plurality of control modules could be operably linked and configured to function together for facilitating various monitoring and control strategies associated with the traction battery pack **14** with the first subpack **62** and the second subpack **64**.
[0042] The control module **46** may include a processor **78** and non-transitory memory **82** for executing various control strategies and modes associated with the method **100**. The processor **78** can be a custom made or commercially available processor, a central processing unit (CPU), or generally any device for executing software instructions. The memory **82** can include a combination of volatile memory elements and nonvolatile memory elements.
[0043] The processor **78** can be operably coupled to the memory **82** and may be configured to execute one or more programs stored in the memory **82** of the control module **74** based on the various inputs received from other devices, such the switches S**1**, S**2**, S**3**, S**4**, and various sensors within and outside of the traction battery pack **14**.
[0044] At a step **108**, after the charging, the method **100** divides the traction battery pack **14** into the first subpack **62** of battery cells **38** and a second subpack **64** of battery cells **38**. The example method **100**, as schematically shown in FIG. **5**, transitions the switching system **70** to divide the traction battery pack **14** in this way. In particular, switches S**1** and S**3** are transitioned to a closed state, and switches S**2** and S**4** are transitioned to an open state. The switches S**1**, S**2**, S**3**, S**4** are maintained in these states when discharging from the first subpack **62**.
[0045] This configuration for the switching system **70** effectively electrically isolates the first subpack **62** from the second subpack **64** and from a load **86**. This configuration for the switching system **70** electrically couples the first subpack **62** to the load **86**. The load **86** can be the electric machine **18**, other high-voltage loads of the electrified vehicle **10**, or both. At a step **112**, the method **100** then discharges the first subpack **62**, but not the second subpack **64**.
[0046] When configured as shown in FIG. **5**, the switching system **70** is considered in a first discharge state. In the first discharge state, the first subpack **62** discharges by providing power to the load **86**. The second subpack **64** is electrically isolated from the load and from the first subpack **62**. The second subpack **64** is thus not discharging, and the first subpack **62** is discharging separately from the second subpack.
[0047] The method **100**, at a step **116**, stops discharging from the first subpack **62** and starts discharging from the second subpack **64**. This can occur in response to a triggering event, such as a state of charge within the first subpack **62** dropping to a threshold state of charge, say ten percent,
[0048] The switching system **70** is reconfigured to make this change. FIG. **6** schematically shows the switching system **70** transitioned so that the second subpack **64** is discharging instead of the first subpack **62**. To make this change, the switches S**1** and S**2** are transitioned or maintained in an open state, and switches S**3** and S**4** are transitioned or maintained in a closed state.
[0049] In some examples, the method **100** includes pre-charging the second subpack **64** to inhibit electrical issues associated with in-rush current when changing from discharging from the first subpack **62** to discharging from the second subpack. The pre-charging may involve a circuit with power resistors to limit in-rush, or a DC/DC converter employed to compensate for imbalances between the first subpack **62** and the second subpack **64**.
[0050] The step **116** can take place when the electrified vehicle **10** is moving during a drive cycle. The triggering event prompting the method **100** to begin discharging from the second subpack **64**

instead of the first subpack **62** can, in some examples, be based on a drive mode. For example, discharging from the first subpack **62** can occur when the electrified vehicle **10** is operating in a front-wheel drive mode, and discharging from the second subpack **64** can occur when the electrified vehicle **10** is operating in an all-wheel drive mode.

[0051] In some examples, the first subpack **62** is designated to power one or more motors of the electrified vehicle **10**, and the second subpack **64** is designated for powered one or more other motors of the electrified vehicle **10**.

[0052] Another example triggering event could include a state of health of the first subpack **62**, the second subpack **64**, or both. Another triggering event could include a thermal energy level within the first subpack **62**, the second subpack **64**, or both. Switching from discharging from the first subpack **62** to discharging from the second subpack **64** can provide time for thermal energy to dissipate from the first subpack **62**.

[0053] Yet another example triggering event could be a range of the electrified vehicle **10** when powered by the first subpack **62**. The switch at the step **116** can occur, for example, when the electrified vehicle **10** has reaches a threshold range for driving when powered by the first subpack **62**. The switch from discharging from the first subpack **62** to the second subpack **64** could occur when a range for the electrified vehicle **10** when powered by the first subpack **62** falls below fifty miles, for example.

[0054] The switch at the step **116** could instead or additionally occur based on a desired state of charge for the first subpack **62** and the second subpack **64** when reaching a destination. For example, if reaching a destination with a substantially equal state of charge in both the first subpack **62** and the second subpack **64** is desired, the switch could occur midway through a journey to the destination. This can help the vehicle **10** to arrive at the destination with both the first subpack **62** and the second subpack **64** at substantially the same temperature, state of charge, etc.

[0055] As can be appreciated, the method **100** can include switching back to a discharging from the first subpack **62** if desired. Further, the method **100** can include discharging concurrently from both the first subpack **62** and the second subpack **64** if, for example, power demands are particularly high. This could be in response to the electrified vehicle **10** climbing a hill, for example.

[0056] Although described in connection with dividing the traction battery pack **14** into the first subpack **62** and the second subpack **64**, it should be understood that other divisions are possible. For example, the traction battery pack **14** could additionally be divided into at least one third subpack. That is, although described as being divided into two subpacks, the traction battery pack **14** could be divided into more than two subpacks.

[0057] In some examples, the control module **74** adjusts cooling of the traction battery pack **14** in response to whether the first subpack **62** or the second subpack **64** is discharging. More coolant could be directed to a thermal exchange plate associated with the first subpack **62** when the first subpack **62** is discharging, for example. In some examples, a cooling system can be sized to cool only the first subpack **62** or the second subpack **64**. Cooling would be, as can be appreciated, directed toward the first subpack **62** when discharging, and then change to the second subpack **64** when discharging.

[0058] Features of the disclosed examples include a traction battery pack dividable into discrete subpacks that are dynamically controlled to combine them during vehicle charge, but separate them during discharge, operating the vehicle from one sub-pack independently during drive. Additional subpacks could be included to, for example, optimize a charge/discharge ratio or to provide compatibility with future voltage standards. Dividing the traction battery pack in this way can lengthen cycle life, particularly for lithium metal cells.

[0059] The preceding description is exemplary rather than limiting in nature. Variations and modifications to the disclosed examples may become apparent to those skilled in the art that do not necessarily depart from the essence of this disclosure. Thus, the scope of protection given to this disclosure can only be determined by studying the following claims.

Claims

1. A traction battery pack operating method, comprising: charging a traction battery pack of an electrified vehicle, the charging including charging a first subpack of battery cells together with a second subpack of battery cells; and discharging the first subpack separately from the second subpack.
2. The method of claim 1, wherein the first subpack of battery cells is a first subpack of battery cells having lithium metal anodes, wherein the second subpack of battery cells is a second subpack of battery cells having lithium metal anodes.
3. The method of claim 1, further comprising, during the discharging, maintaining at least one switch in an open state to electrically isolate the first subpack from the second subpack.
4. The method of claim 1, further comprising, when discharging the first subpack separately from the second subpack, providing electrical power from the first subpack to a load, and electrically isolating the second subpack from the load.
5. The method of claim 4, further comprising transitioning at least one switch to electrically isolate the second subpack from the load.
6. The method of claim 4, wherein the load is at least one motor of the electrified vehicle.
7. The method of claim 1, further comprising switching from discharging the first subpack separately from the second subpack, to discharging the second subpack separately from the first subpack.
8. The method of claim 7, further comprising pre-charging the second subpack during the switching.
9. The method of claim 7, wherein the discharging is during a drive cycle.
10. The method of claim 7, further comprising switching based on a state of health in the first subpack, the second subpack, or both.
11. The method of claim 7, further comprising switching based on a drive mode of the electrified vehicle.
12. The method of claim 7, further comprising switching based on a range for the electrified vehicle when discharging from the first subpack reaching a threshold range.
13. The method of claim 1, wherein the discharging the first subpack separately from the second subpack comprises discharging the first subpack to a load without discharging the second subpack to the load.
14. The method of claim 1, further comprising dividing the traction battery pack into the first subpack of battery cells, and the second subpack of battery cells, wherein dividing the traction battery pack into the first subpack and the second subpack comprises electrically isolating the second subpack from both the first subpack and a load.
15. The method of claim 1, wherein the traction battery pack is an 800 Volt traction battery pack, wherein the first subpack is 400 Volts and the second subpack is 400 Volts.
16. The method of claim 1, further comprising dividing the traction battery pack into the first subpack of battery cells, and the second subpack of battery cells, wherein the dividing further comprises dividing the traction battery pack into at least one third subpack, and the discharging is a discharging of the first subpack without discharging the second subpack or the at least one third subpack.
17. The method of claim 1, wherein the first subpack and the second subpack are both held within a battery enclosure.
18. A traction battery assembly, comprising: a traction battery pack that powers a load; and a switching system that is configured to transition between a charging state, a first discharging state, and a second discharging state, when the switching system is in charging state, a first subpack of battery cells in the traction battery pack and a second subpack of battery cells in the traction battery

pack are both configured to be charged, when the switching system is in first discharging state, the first subpack is electrically connected to the load while the second subpack is electrically isolated from the load, when the switching system is in the second discharging state, the second subpack is electrically connected to the load while the first subpack is electrically isolated from the load.

19. The traction battery assembly of claim 18, wherein the switching system electrically isolates the first subpack of battery cells from the second subpack of battery when the switching system is in the first discharging state and when the switching system is in the second discharging state.

20. The traction battery assembly of claim 18, wherein the first subpack of battery cells is a first subpack of battery cells having lithium metal anodes, wherein the second subpack of battery cells is a second subpack of battery cells having lithium metal anodes.